

LAB

Topic 8: Indexing



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Indexing

An index helps SQL Server locate rows without scanning the entire table. It can improve the performance of queries using WHERE, JOIN, and ORDER BY.

However, indexes require additional storage and must be updated when data is inserted, deleted, or modified. Therefore, indexes should be created only when they support useful queries.

General Syntax

```
CREATE [UNIQUE] [CLUSTERED | NONCLUSTERED] INDEX index_name  
ON table_name (column1, column2, ...)  
INCLUDE (column3, column4, ...)  
WHERE filter_condition;
```

Common index types

Clustered index

A clustered index organizes the table rows according to the index key. A table can have only one clustered index.

```
CREATE CLUSTERED INDEX IX_Student_ID  
ON Student_IndexDemo(student_id);
```

Nonclustered index

A nonclustered index is stored separately from the table data. A table can have multiple nonclustered indexes.

```
CREATE NONCLUSTERED INDEX IX_Student_Department  
ON Student_IndexDemo(department_id);
```

Unique index

```
CREATE UNIQUE INDEX IX_Student_Name  
ON Student_IndexDemo(student_name);
```

Composite index

```
CREATE NONCLUSTERED INDEX IX_Student_Department_Class  
ON Student_IndexDemo(department_id, class);
```

Index with included columns

```
CREATE NONCLUSTERED INDEX IX_Student_Department_Covering  
ON Student_IndexDemo(department_id)
```

```
INCLUDE (student_name, class);
```

Filtered index

```
CREATE NONCLUSTERED INDEX IX_Student_HasDepartment  
ON Student_IndexDemo(department_id)  
WHERE department_id IS NOT NULL;
```

Removing an index

```
DROP INDEX IX_Student_Department  
ON Student_IndexDemo;
```